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Filtration system for a water drainage system

Abstract

A downspout drain filter includes a housing and a filter media. The housing defines an internal cavity, an inlet to the internal cavity along a top of the housing, and an outlet from the internal cavity along a bottom of the housing. The filter media is disposed within the internal cavity. The filter media divides the internal cavity into an upper chamber and a lower chamber. The filter media is configured to trap debris in the upper chamber from fluid flowing downward through the internal cavity. The filter media is permeable such that the fluid flows from the upper chamber to the lower chamber.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. patent application Ser. No. 17/704,439 filed on Mar. 25, 2022, now U.S. Pat. No. 11,471,796, issued Oct. 18, 2022, the disclosure of which is hereby incorporated in its entirety by reference herein.

TECHNICAL FIELD

(1) The present disclosure relates to water drainage systems, particularly to water drainage systems for buildings and houses.

BACKGROUND

(2) Runoff water may be directed away from buildings and houses via water drainages systems

SUMMARY

(3) A downspout drain filter includes a housing and a filter plate. The housing defines an internal cavity, an inlet to the internal cavity along a top of the housing, and an outlet from the internal cavity along a bottom of the housing. The filter plate is disposed within the internal cavity. The filter plate divides the internal cavity into an upper chamber and a lower chamber. The filter plate is configured to trap debris in the upper chamber from fluid flowing downward through the internal cavity. The filter plate is permeable such that the fluid flows from the upper chamber to the lower chamber. The filter plate has a removable central portion that provides access to the lower chamber from the upper chamber when removed.

(4) A downspout drain filter includes a housing and a filter media. The housing defines an internal cavity, an inlet to the internal cavity along a top of the housing, and an outlet from the internal cavity along a bottom of the housing. The filter media is disposed within the internal cavity. The filter media divides the internal cavity into an upper chamber and a lower chamber. The filter media is configured to trap debris in the upper chamber from fluid flowing downward through the internal cavity. The filter media is permeable such that the fluid flows from the upper chamber to the lower chamber. The housing further defines an overflow opening along the top of the housing. The overflow opening is configured to direct the fluid out of the internal cavity in response to the filter media becoming clogged and the fluid filling the upper chamber.

(5) A downspout drain filter includes a housing, The housing includes a top end, a bottom end, a front panel, a back panel, and a pair of opposing side panels, and at least one sloped surface. The front panel, back panel, and side panels collectively define an internal cavity. Each of the front, back, and side panels are disposed between the top end and the bottom end. The top end defines an inlet to the internal cavity. The bottom end defines an outlet from the internal cavity. The at least one sloped surface extends into the internal cavity such that the internal cavity tapers in a direction that extends from the top end to the bottom end of the housing along the at least one sloped surface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front view of a water drainage systems for a building or house;
- (2) FIG. 2 is an isometric front view of a downspout drain filter for the water drainage system;
- (3) FIG. 3 is a cross-sectional view taken along line 3-3 in FIG. 2;
- (4) FIG. 4 is a cross-sectional view taken along line 4-4 in FIG. 2;
- (5) FIG. 5 is a cross-sectional view taken along line 5-5 in FIG. 2;
- (6) FIG. 6 is an isometric front view of a filter plate for the downspout drain filter in an assembled state;
- (7) FIG. 7 is an exploded view of the filter plate in a disassembled state;
- (8) FIG. 8 is a cross-sectional view of an alternative engagement arrangement between the housing and access panel of downspout drain filter;
- (9) FIG. 9 is an isometric front view of an alternative embodiment of the downspout drain filter for the water drainage system;
- (10) FIG. 10 is a back view of the alternative embodiment of the downspout drain filter;
- (11) FIG. 11 is a front view of the alternative embodiment of the downspout drain filter with the access panel removed;

(12) FIG. 12 is a top view of the alternative embodiment of the downspout drain filter with the access panel and the filter plate removed;

(13) FIG. 13 is a top view of the alternative embodiment of the downspout drain filter with the filter plate disposed therein; and

(14) FIGS. 14 and 15 are isometric cross-sections side views of the alternative embodiment of the downspout drain filter taken along line 14-14 in FIG. 9.

DETAILED DESCRIPTION

(15) Embodiments of the present disclosure are described herein. It is to be understood, however, that the disclosed embodiments are merely examples and other embodiments may take various and alternative forms. The figures are not necessarily to scale; some features could be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the embodiments. As those of ordinary skill in the art will understand, various features illustrated and described with reference to any one of the figures may be combined with features illustrated in one or more other figures to produce embodiments that are not explicitly illustrated or described. The combinations of features illustrated provide representative embodiments for typical applications. Various combinations and modifications of the features consistent with the teachings of this disclosure, however, could be desired for particular applications or implementations.

(16) Referring to FIG. 1, a water drainage system 10 is illustrated. The water drainage system 10 may specifically be designed to direct fluid or water away from a house or building in order to prevent runoff water from the house or building from flooding the lower level (e.g., the basement) of the house or building. It should be understood, however, that the water drainage system 10 may be utilized in other scenarios and should not be construed as limited to a water drainage system that is configured to direct water away from a house or building.

(17) As illustrated in FIG. 1, the water drainage system 10 includes tubes, pipes, or conduits 12 that are configured to direct water away from a house or building 14. More specifically, the conduits 12 may be configured to direct water away from a gutter or eavestrough 16 that is disposed along edges of a roof of the building 14. The eavestrough 16 is configured to collect runoff water that cascades down the roof of the building 14 during rainstorms, from melting snow, or from any other source. The conduits 12 are configured to direct the water away from the eavestrough 16 and away from the building 14. More specifically, the conduits 12 may be configured to direct the water toward a storm drain.

(18) Catch basins of storm drains (i.e., the portions of storm drains that are the inlets to a storm drain system that are typically vertically oriented) are usually disposed along an edge of the pavement of a road or street and adjacent to a curb of the road or street. The pavement typically has a slight slope to direct the runoff water from the road or street into the catch basin. The curb may also be positioned above the catch basin and is sloped downward to direct runoff water from the ground that is adjacent to the catch basin toward the catch basin.

(19) The tubes, pipes, or conduits 12 of the water drainage system 10 may include a downspout 18 that is configured to direct water downward from the eavestrough 16 to a drainage tube, pipe, or conduit 20. Stated in other terms, the downspout 18 establishes fluid communication between the eavestrough 16 and the drainage conduit 20. The drainage conduit 20 in turn may be configured to direct water from the downspout 18 to an outlet end 22 of the water drainage system 10. The drainage conduit 20 may include a slight downward slope extending in a direction from the downspout 18 to the outlet end 22 of the water drainage system 10 such that water flows from the downspout 18 to the outlet end 22 of the water drainage system 10. The drainage conduit 20 may include a bend that connects the drainage conduit 20 to the downspout 18. Alternatively, an elbow tube, pipe, or conduit 24 may connect the drainage conduit 20 to the downspout 18. The drainage conduit 20 and the first elbow conduit 24 may collectively be referred to as the drainage tube, pipe,

or conduit.

(20) The water drainage system **10** may include a filter **26** that is disposed between the downspout **18** and the drainage conduit **20**. The filter **26** may be in fluid communication with both the downspout **18** and the drainage conduit **20**. The filter **26** may also establish fluid communication between the downspout **18** and the drainage conduit **20**. The filter **26** is configured to filter debris from fluid or water flowing from the downspout **18** to the drainage conduit **20**. The filter **26** may be oriented vertically along or adjacent to the downspout **18**, which may also be oriented vertically. In some scenarios, the filter **26** may divide the downspout **18** into an upper and a lower conduit. The filter **26** may be referred to as a downspout drain filter.

(21) The drainage conduit **20** and the first elbow conduit **24** may be fully or partially disposed or buried underground, which may be for aesthetic purposes or may be to prevent placing an obstacle on an area of ground that may require maintenance (e.g., the area of ground may include grass that needs to be mowed on a regular basis) or that may have a significant amount of foot traffic. Alternatively, the drainage conduit **20** and the first elbow conduit **24** may be disposed above ground. The outlet end **22** of the water drainage system **10** may be connected directly to a storm drain or more specifically to a catch basin of a storm drain. Alternately, the outlet end **22** of the water drainage system **10** may include an opening that allows water to flow out of the water drainage system **10**. Such an opening in the outlet end **22** may be positioned near an opening to a catch basin such that the water flowing out of the opening at the outlet end **22** is directed toward the opening to the catch basin.

(22) As previously stated, the conduits **12** may be configured to direct water away from any source, and this disclosure should not be construed as limited to the layout of conduits **12** in FIG. **1**. For example, the elbow conduit **24** is illustrated as a 90° elbow (i.e., an elbow pipe that is bent at about a 90° angle), but may be reconfigured as an elbow that is bent at any desirable angle (e.g., 60°, 45°, and 22.5°, etc.).

(23) Referring to FIGS. **2-3**, the filter **26** is illustrated in further detail. The filter **26** includes a housing **28**. The housing defines an internal chamber or cavity **30**, a first opening **32** to the internal cavity **30**, a second opening **34** to the internal cavity **30**, and a third opening **36** to the internal cavity **30**. The first opening **32** to the internal cavity **30** is defined along a top or a top end **38** of the housing **28**. The second opening **34** to the internal cavity **30** is defined along a bottom or a bottom end **40** of the housing **28**. The third opening **36** to the internal cavity **30** is defined along front face **42** of the housing **28**. The first opening **32** is an inlet to the internal cavity **30**. The first opening **32** establishes fluid communication between the downspout **18** and the internal cavity **30**. The second opening **34** is an outlet from the internal cavity **30**. The second opening **34** establishes fluid communication between the internal cavity **30** and the drainage conduit **20**.

(24) A filter media or filter plate **44** is disposed within the internal cavity **30**. The filter plate **44** may more specifically be a grated plate. The filter plate **44** may be disposed within the internal cavity **30** at an angle relative to a vertical direction **46**. The filter plate **44** may divide the internal cavity **30** into an upper chamber **48** and a lower chamber **50**. The filter plate **44** is configured to trap debris, such as leaves, in the upper chamber **48** from fluid or water flowing downward through the internal cavity **30**, while also being permeable such that the fluid or water flows through the filter plate **44** from the upper chamber **48** to the lower chamber **50**.

(25) A door or access panel **52** is secured to the housing **28** and over the third opening **36**. The access panel **52** is configured to provide access to the upper chamber **48** for removal of debris trapped within the upper chamber **48**. The housing **28** may further define an overflow opening **54** that is disposed above the access panel **52** along the top end **38** of the housing **28**. The overflow opening **54** may be grated to prevent debris from entering into the internal cavity **30** via the overflow opening **54**. If grated, the overflow opening **54** may more specifically comprise an array or a series of adjacent openings. The overflow opening **54** is configured to direct the fluid or water out of the internal cavity **30** in response to the filter plate **44** becoming clogged and fluid or water

filling the upper chamber **48**. The overflow opening **54** may be in direct fluid communication with the upper chamber **48**.

(26) The overflow opening **54** may be defined along the top of the upper chamber **48** so that under most conditions fluid or water will not flow out of the overflow opening but will flow downward via gravity through the filter plate **44** and out of the internal cavity **30** via the outlet or second opening **34**. Only when the rate of fluid or water flowing into the upper chamber **48** via the inlet or first opening **32** exceeds the rate of fluid or water flowing through the filter plate **44** for a period of time, will the fluid or water flow out of the overflow opening **54**. The fluid or water flowing out of the overflow opening **54** may then function as a signal that debris, such as leaves, has clogged the filter plate **44**. Such a signal may act to notify an individual that is responsible for maintaining the filter **26** that debris needs to be removed from the internal cavity **30**, or more specifically that debris needs to be removed from the upper chamber **48** of the internal cavity **30**.

(27) The overflow opening **54** may be positioned adjacent to and above the front face **42**. The front face **42** and overflow opening **54** may be positioned to face outward from a building or house that the downspout **18** and filter **26** are secure to, so that it is easy to observe fluid or water flowing out of the overflow opening **54**. The overflow opening **54** may also be orientated at an angle relative to the front face **42** of the housing **28**, such that fluid or water flowing out the overflow opening **54** may be directed partially upward and partially outward from the filter **26**. This further increases the ease of observing fluid or water flowing out of the overflow opening **54**, since the fluid or water is being directed outward from the filter **26** and not simply downward where the fluid or water may trace along the outer surface of the filter **26** or any adjacent conduit that is disposed below the filter **26**, which would be more difficult to observe particularly from a distance.

(28) Referring now to FIGS. 2-5, the top end **38** and the bottom end **40** of the housing **28** are described in further detail. The top end **38** of the housing **28** may be a separate component from a remainder **56** of the housing **28**. The top end **38** of the housing **28** may be secured to the remainder **56** of the housing **28** via fasteners, clips, snaps, etc. The top end **38** of the housing **28** and the remainder **56** of the housing **28** may also define slots **58**, that are configured to receive portions of the filter plate **44** and the access panel **52**. The sections of the top end **38** of the housing **28** and the remainder **56** of the housing **28** that are adjacent to the slot **58** may function to trap the portions of the filter plate **44** and the access panel **52** within the slots **58** to retain the filter plate **44** and the access panel **52** in desired positions (e.g., the positions depicted in FIGS. 2-3). The filter plate **44** and the access panel **52** may be flexible so that the filter plate **44** and the access panel **52** may be manipulated into and out of the respective desired positions during installation onto or removal from the housing **28**. The filter plate **44** and the access panel **52** may also be sufficiently rigid to remain in the respective desired position when no external force is applied.

(29) The top end **38** of the housing **28** includes a step **60**. A cross-sectional area of the first opening **32** is decreased from a first value **62** to a second value **64** along the step **60** and in a direction that extends away from the internal cavity **30**. For example, the portion of the first opening **32** where the cross-sectional area is equal to the first value **62** may be sized to receive a four by three inch rectangular tube, conduit, or pipe, while the portion of the first opening **32** where the cross-sectional area is equal to the second value **64** may be sized to receive a three by two inch rectangular tube, conduit, or pipe. In the event the larger-sized connection (e.g., the portion of the top end **38** where the cross-sectional area of the first opening **32** is equal to the second value **64**) is required for connecting the housing **28** to the downspout **18**, the smaller-sized connection (e.g., the portion of the top end **38** where the cross-sectional area of the first opening **32** is equal to the first value **62**) may be cut off from the remainder of the housing **28**. For example, the portion of the top end **38** that includes the smaller-sized connection may be cut off from the remainder of the housing **28** by cutting the housing **28** along dashed line **65**.

(30) The bottom end **40** of the housing **28** may be sized for connecting to a circular tube, conduit, or pipe. The bottom end **40** of the housing **28** may include clips, snaps, or tabs **66** to connect to a

corrugated tube, conduit, or pipe. In the event the bottom end **40** is to be connected to a non-corrugated tube, conduit, or pipe, a portion of the bottom end **40** that includes the clips, snaps, or tabs **66** may be cut off from the remainder of the housing **28**. For example, the portion of the bottom end **40** that includes the clips, snaps, or tabs **66** may be cut off from the remainder of the housing **28** by cutting the housing **28** along dashed line **68**.

(31) Referring now to FIGS. **3** and **6-7**, the filter plate **44** is illustrated in further detail. The filter plate **44** includes a removable central portion **70** that provides access to the lower chamber **50** from the upper chamber **48** when removed. The removable central portion **70** may be removed so that a hose, jet, heat cable, etc. may be placed into the drainage conduit **20** that is positioned below the bottom end **40** of the housing **28**. A hose or jet may be used to clean out the drainage conduit **20**. A heat cable may be used to prevent water from freezing within the drainage conduit **20**, particularly during the winter months, which could cause damage to the conduit.

(32) The removable central portion **70** of the filter plate **44** is configured to lock and unlock to and from a remainder **72** of the filter plate **44** via rotation relative to the remainder **72** of the filter plate **44**. The remainder **72** of the filter plate **44** may be referred to as the main portion of the filter plate **44**. The removable central portion **70** of the filter plate **44** includes a first set of tabs **74**. The remainder **72** of the filter plate includes a second set of tabs **76**. The first set of tabs **74** are configured to align with the second tabs **76** to lock the removable central portion **70** of the filter plate **44** to the remainder **72** of the filter plate **44**. The remainder **72** of the filter plate **44** defines slots **78**. The first set of tabs **74** are configured to align with the slots **78** to unlock the removable central portion **70** of the filter plate **44** from the remainder **72** of the filter plate. The filter plate **44** may include ribs **80** that extending upward from an upper surface **82** of the filter plate **44**. The ribs **80** are configured to separate debris from the upper surface **82** of the filter plate **44** to facilitate flow of fluid or water through the filter plate **44**. More specifically, the ribs **80** are configured to lift the debris off of the upper surface **82** of the filter plate **44** so the openings **84** defined by the grating of the filter plate **44** are separated from the debris so that the fluid or water may flow through the openings **84**.

(33) Referring to FIG. **8**, a cross-sectional view of an alternative engagement arrangement between the housing **28** and the access panel **52** of the filter **26** is illustrated. The access panel **52** includes a protrusion **86** extending downward from a bottom end or bottom surface **88** of the access panel **52**. The housing **28** defines a slot **90** along a ledge or shelf **92** upon which the access panel **52** is configured to rest when secured to the housing **28**. The protrusion **86** is disposed within the slot **90** to further secure the access panel **52** to the housing **28**.

(34) Referring to FIGS. **9-15**, an alternative embodiment of the filter **126** is illustrated. Unless otherwise stated herein, the filter **126** should be construed to include all of the same elements, structure, and functionally as filter **26** unless otherwise stated herein. The filter **126** includes a housing **128**. The housing **128** defines an internal chamber or cavity **130**, a first opening **132** to the internal cavity **130**, a second opening **134** to the internal cavity **130**, and a third opening **136** to the internal cavity **130**. The first opening **132** to the internal cavity **130** is defined by or along a top or a top end **138** of the housing **128**. The second opening **134** to the internal cavity **130** is defined by or along a bottom or a bottom end **140** of the housing **128**. The third opening **136** to the internal cavity **130** is defined along front face **142** of the housing **128**. The first opening **132** is an inlet to the internal cavity **130**. The first opening **132** establishes fluid communication between the downspout **18** and the internal cavity **130** when the filter **126** is secured the downspout **18**. The second opening **134** is an outlet from the internal cavity **130**. The second opening **134** establishes fluid communication between the internal cavity **130** and the drainage conduit **20** when the filter **126** is secured the drainage conduit **20** via elbow conduit **24**.

(35) A filter media or filter plate **144** is disposed within the internal cavity **130**. The filter plate **144** may more specifically be a grated plate. The filter plate **144** may be disposed within the internal cavity **130** at an angle relative to a vertical direction **146**. The filter plate **144** may divide the internal cavity **130** into an upper chamber **148** and a lower chamber **150**. The filter plate **144** is

configured to trap debris, such as leaves, in the upper chamber **148** from fluid or water flowing downward through the internal cavity **130**, while also being permeable such that the fluid or water flows through the filter plate **144** from the upper chamber **148** to the lower chamber **150**.

(36) A door or access panel **152** is secured to the housing **128** and over the third opening **136**. The access panel **152** is configured to provide access to the upper chamber **148** for removal of debris trapped within the upper chamber **148**. The housing **128** may further define an overflow opening **154** that is disposed above the access panel **152** along the top end **138** of the housing **128**. The overflow opening **154** may be grated to prevent debris from entering into the internal cavity **130** via the overflow opening **154**. If grated, the overflow opening **154** may more specifically comprise an array or a series of adjacent openings. The overflow opening **154** is configured to direct the fluid or water out of the internal cavity **130** in response to the filter plate **144** becoming clogged and fluid or water filling the upper chamber **148**. The overflow opening **154** may be in direct fluid communication with the upper chamber **148**.

(37) The overflow opening **154** may be defined along the top of the upper chamber **148** so that under most conditions fluid or water will not flow out of the overflow opening **154** but will flow downward via gravity through the filter plate **144** and out of the internal cavity **130** via the outlet or second opening **134**. Only when the rate of fluid or water flowing into the upper chamber **148** via the inlet or first opening **132** exceeds the rate of fluid or water flowing through the filter plate **144** for a period of time, will the fluid or water flow out of the overflow opening **154**. The fluid or water flowing out of the overflow opening **154** may then function as a signal that debris, such as leaves, has clogged the filter plate **144**. Such a signal may act to notify an individual that is responsible for maintaining the filter **126** that debris needs to be removed from the internal cavity **130**, or more specifically that debris needs to be removed from the upper chamber **148** of the internal cavity **130**.

(38) The overflow opening **154** may be positioned adjacent to and above the front face **142**. The front face **142** and overflow opening **154** may be positioned to face outward from a building or house that the downspout **18** and filter **126** are secure to, so that it is easy to observe fluid or water flowing out of the overflow opening **154**. The overflow opening **154** may also be orientated at an angle relative to the front face **142** of the housing **128**, such that fluid or water flowing out the overflow opening **154** may be directed partially upward and partially outward from the filter **126**. This further increases the ease of observing fluid or water flowing out of the overflow opening **154**, since the fluid or water is being directed outward from the filter **126** and not simply downward where the fluid or water may trace along the outer surface of the filter **126** or any adjacent conduit that is disposed below the filter **126**, which would be more difficult to observe particularly from a distance.

(39) The top end **138** of the housing **128** may be a separate component from a remainder **156** of the housing **128**. The top end **138** of the housing **128** may be secured to the remainder **156** of the housing **128** via fasteners, clips, snaps, etc. The top end **138** of the housing **128** and the remainder **156** of the housing **128** may also define a slot **158**, that is configured to receive a portion of the access panel **152**. The sections of the top end **138** of the housing **128** and the remainder **156** of the housing **128** that are adjacent to the slot **158** may function to trap the portions of the access panel **152** within the slot **158** to retain the access panel **152** in a desired position (e.g., the position depicted in FIGS. **9** and **14-15**). The filter plate **144** and the access panel **152** may be flexible so that the filter plate **144** and the access panel **152** may be manipulated into and out of the respective desired positions during installation onto or removal from the housing **128**. The filter plate **144** and the access panel **152** may also be sufficiently rigid to remain in the respective desired positioned when no external force is applied.

(40) The top end **138** of the housing **128** may include a step **160**. A cross-sectional area of the first opening **32** is decreased from a first value to a second value along the step **160** and in a direction that extends away from the internal cavity **130**. For example, the portion of the first opening **132**

where the cross-sectional area is equal to the first value may be sized to receive a four by three inch rectangular tube, conduit, or pipe, while the portion of the first opening **132** where the cross-sectional area is equal to the second value may be sized to receive a three by two inch rectangular tube, conduit, or pipe. In the event the larger-sized connection (e.g., the portion of the top end **138** where the cross-sectional area of the first opening **132** is equal to the second value) is required for connecting the housing **128** to the downspout **118**, the smaller-sized connection (e.g., the portion of the top end **138** where the cross-sectional area of the first opening **132** is equal to the first value) may be cut off from the remainder of the housing **128**. For example, the portion of the top end **138** that includes the smaller-sized connection may be cut off from the remainder of the housing **128** by cutting the housing **128** along dashed line **165**.

(41) The bottom end **140** of the housing **128** may be sized for connecting to a circular tube, conduit, or pipe. The bottom end **140** of the housing **128** may include clips, snaps, or tabs **166** to connect to a corrugated tube, conduit, or pipe. In the event the bottom end **140** is to be connected to a non-corrugated tube, conduit, or pipe, a portion of the bottom end **140** that includes the clips, snaps, or tabs **166** may be cut off from the remainder of the housing **128**. For example, the portion of the bottom end **140** that includes the clips, snaps, or tabs **166** may be cut off from the remainder of the housing **128** by cutting the housing **128** along dashed line **168**.

(42) The filter plate **144** includes a removable central portion **170** that provides access to the lower chamber **150** from the upper chamber **148** when removed. The removable central portion **170** may be removed so that a hose, jet, heat cable, etc. may be placed into the drainage conduit **20** that is positioned below the bottom end **140** of the housing **128**. A hose or jet may be used to clean out the drainage conduit **20**. A heat cable may be used to prevent water from freezing within the drainage conduit **20**, particularly during the winter months, which could cause damage to the conduit.

(43) The removable central portion **170** of the filter plate **144** is configured to lock and unlock to and from a remainder of the filter plate **144** via rotation relative the remainder of the filter plate **144**. The remainder of the filter plate **44** may be referred to as the main portion of the filter plate **144**. The removable central portion **170** of the filter plate **144** includes a first set of tabs. The remainder of the filter plate includes a second set of tabs. The first set of tabs are configured to align with the second tabs to lock the removable central portion **170** of the filter plate **144** to the remainder of the filter plate **144**. The remainder of the filter plate **144** defines slots. The first set of tabs are configured to align with the slots to unlock the removable central portion **170** of the filter plate **144** from the remainder of the filter plate. The filter plate **144** may include ribs that extending upward from an upper surface of the filter plate **144**. The ribs are configured to separate debris from the upper surface of the filter plate **144** to facilitate flow of fluid or water through the filter plate **144**. More specifically, the ribs are configured to lift the debris off of the upper surface of the filter plate **144** so the openings defined by the grating of the filter plate **144** are separated from the debris so that the fluid or water may flow through the openings.

(44) The housing **128** also includes a front panel **172**, a back panel **174**, and a pair of opposing spaced-apart side panels **176**. The front panel **172**, back panel **174**, and side panels **176** may collectively define the internal cavity **130**. The pair of opposing spaced-apart side panels **176** may also be referred to as the first and second side panels. The front panel **172**, back panel **174**, and side panels **176** are disposed between the top end **138** and the bottom end **138**. The housing **128** also includes at least one sloped surface **178** that extends into the internal cavity **130** and partially defines the internal cavity **130** such that the internal cavity tapers **130** in a direction **180** that extends from the top end **138** to the bottom end **140** of the housing **128** along the at least one sloped surface **178**. Direction **180** may correspond to vertical direction **146**. The at least one sloped surface **178** operates to channel the water that is flowing downward through the housing **128** toward the second opening **134** so that the water is directed toward the drainage conduit **20**. The at least one sloped surface **178** also operates to channel water so that it is not directed to the access panel **152** in order to reduce or eliminate water leakage from the housing **128** via gaps between the

access panel **152** and the housing **128** along the third opening **136**.

(45) More specifically, the at least one sloped surface **178** may include two sloped surfaces. The two sloped surfaces may be referred to as first and second sloped surfaces. Each of the sloped surfaces **178** may extend between the back panel **174** and one of the side panels **176**. Even more specifically, a first of the sloped surfaces **178** may extend between the back panel **174** and the first of the side panels **176** while a second of the sloped surfaces **178** may extend between the back panel **174** and the second of the side panels **176**.

(46) The front panel **172** may define the third opening **136**, which may be referred to as an access opening. The access panel **152** is disposed over the third opening **136** and may be removed to provide access to the internal cavity **130** via the third opening so that debris may be removed from the internal cavity **130**. The access panel **152** has a secondary sloped surface **182** that extends into the internal cavity **130** and partially defines the internal cavity **130** such that the internal cavity **130** tapers in the direction **180** that extends from the top end **138** to the bottom end **140** of the housing **128** along the secondary sloped surface **182**. The secondary sloped surface **182** further operates to channel the water that is flowing downward through the housing **128** toward the second opening **134** so that the water is directed toward the drainage conduit **20**. The secondary sloped surface **182** also operates to channel water away from the access panel **152** to reduce or eliminate water leakage from the housing **128** via gaps between the access panel **152** and the housing **128** along the third opening **136**.

(47) The housing **128** also includes a frustoconical section **184** that is disposed below the at least one sloped surface **178** and the secondary sloped surface **182**. An internal surface **185** of the frustoconical section **184** is sloped or funnel-shaped, extends into the internal cavity **130**, and partially defines the internal cavity **130** such that the internal cavity **130** further tapers in the direction **180** that extends from the top end **138** to the bottom end **140** of the housing **128** along the internal surface **185** of the frustoconical section **184**. The internal surface **185** of the frustoconical section **184** further operates to channel the water that is flowing downward through the housing **128** toward the second opening **134** so that the water is directed toward the drainage conduit **20**.

(48) The filter plate **144** may be more specifically disposed within the internal cavity **130** between the front panel **172** and the back panel **174**. The filter plate **144** may also be positioned at an angle within the internal cavity **130** such that an upper end of the filter plate **144** is adjacent to the back panel **174** while a lower end of the filter plate **144** is adjacent to the front panel **172**. The filter plate **144** may include offset sections **186** that are disposed adjacent to and provide clearance for the sloped surfaces **178**. More specifically, the filter plate **144** may include first and second offset sections **186** that are offset upward along the top surface of the filter plate **144**, and are disposed adjacent to and provide clearance for the first and second sloped surfaces **178**, respectively. The offset sections **186** may also be grated so debris may be collected along on the top sides of the offset sections **186** while fluid or water may flow through the openings defined by the grating along the offset sections **186** of the filter plate **144**.

(49) The top end **138** of the housing **128** may engage a top end of the filter plate **144** along an angled ledge **188**. The housing **128** may further include tabs **190** that engage the bottom end of the filter plate **144**. Collectively, the angled ledge **188** and the tabs **190** operate to retain the filter plate within the internal cavity **130**.

(50) It should be understood that the designations of first, second, third, fourth, etc. for any component, state, or condition described herein may be rearranged in the claims so that they are in chronological order with respect to the claims. Furthermore, it should be understood that any component, state, or condition described herein that does not have a numerical designation may be given a designation of first, second, third, fourth, etc. in the claims if one or more of the specific component, state, or condition are claimed.

(51) The words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the

disclosure. As previously described, the features of various embodiments may be combined to form further embodiments that may not be explicitly described or illustrated. While various embodiments could have been described as providing advantages or being preferred over other embodiments or prior art implementations with respect to one or more desired characteristics, those of ordinary skill in the art recognize that one or more features or characteristics may be compromised to achieve desired overall system attributes, which depend on the specific application and implementation. As such, embodiments described as less desirable than other embodiments or prior art implementations with respect to one or more characteristics are not outside the scope of the disclosure and may be desirable for particular applications.

Claims

1. A downspout drain filter comprising: a housing defining (i) an internal cavity, (ii) an inlet to the internal cavity along a top of the housing, and (iii) an outlet from the internal cavity along a bottom of the housing, wherein the housing includes (a) front and back panels that are disposed between the top and bottom of the housing, (b) first and second side panels extending between the front and back panels, and (c) first and second sloped surfaces extending between the back panel and the first and second side panels, respectively, wherein the first and second sloped surfaces extend into the internal cavity such that the internal cavity tapers in the direction that extends from the top to the bottom of the housing along the first and second sloped surfaces; and a filter plate (i) disposed within the internal cavity, (ii) dividing the internal cavity into an upper chamber and a lower chamber, (iii) configured to trap debris in the upper chamber from fluid flowing downward through the internal cavity, (iv) permeable such that the fluid flows from the upper chamber to the lower chamber, and (v) having a removable central portion that provides access to the lower chamber from the upper chamber when removed.
2. The downspout drain filter of claim 1, wherein the removable central portion of the filter plate is configured to lock and unlock to and from a remainder of the filter plate via rotation relative the remainder of the filter plate.
3. The downspout drain filter of claim 2, wherein (i) the removable central portion of the filter plate includes a first set of tabs, (ii) the remainder of the filter plate includes a second set of tabs, and (iii) the first set of tabs are configured to align with the second set of tabs to lock the removable central portion of the filter plate to the remainder of the filter plate.
4. The downspout drain filter of claim 3, wherein (i) the remainder of the filter plate defines slots and (ii) the first set of tabs are configured to align with the slots to unlock the removable central portion of the filter plate from the remainder of the filter plate.
5. The downspout drain filter of claim 1, wherein the filter plate includes ribs extending upward from an upper surface of the filter plate, and wherein the ribs are configured to separate the debris from the upper surface of the filter plate to facilitate flow of the fluid through the filter plate.
6. The downspout drain filter of claim 1, wherein (i) the front panel defines an access opening, (ii) a door is disposed over the access opening, and (iii) the door has a third sloped surface that extends into the internal cavity such that the internal cavity tapers in a direction that extends from the top to the bottom of the housing along the third sloped surface.
7. The downspout drain filter of claim 6, wherein the back panel is opposite of the front panel.
8. A downspout drain filter comprising: a housing defining (i) an internal cavity, (ii) an inlet to the internal cavity along a top of the housing, and (iii) an outlet from the internal cavity along a bottom of the housing, wherein the housing includes (a) a back panel disposed between the top and bottom of the housing, (b) first and second side panels extending between the back panel and a front panel, and (c) first and second sloped surfaces extending between the back panel and the first and second side panels, respectively, wherein the first and second sloped surfaces extend into the internal cavity such that the internal cavity tapers in the direction that extends from the top to the bottom of

the housing along the first and second sloped surfaces; and a filter media (i) disposed within the internal cavity, (ii) dividing the internal cavity into an upper chamber and a lower chamber, and (iii) configured to trap debris in the upper chamber from fluid flowing downward through the internal cavity, and (iv) permeable such that the fluid flows from the upper chamber to the lower chamber, wherein the housing further defines an overflow opening along the top of the housing, and wherein the overflow opening is configured to direct the fluid out of the internal cavity in response to (a) the filter media becoming clogged and (b) the fluid filling the upper chamber.

9. The downspout drain filter of claim 8, wherein the overflow opening is orientated at an angle relative to a front face of the housing.

10. The downspout drain filter of claim 9, wherein the overflow opening is adjacent to and above the front face.

11. The downspout drain filter of claim 8, wherein the overflow opening is grated.

12. The downspout drain filter of claim 8, wherein the filter media includes a removable central portion that provides access to the lower chamber from the upper chamber when removed.

13. The downspout drain filter of claim 8, wherein (i) the front panel disposed between the top and bottom of the housing, (ii) the front panel defines an access opening, (iii) a door is disposed over the access opening, and (iv) the door has a third sloped surface that extends into the internal cavity such that the internal cavity tapers in a direction that extends from the top to the bottom of the housing along the third sloped surface.

14. The downspout drain filter of claim 13, wherein the back panel is opposite of the front panel.
